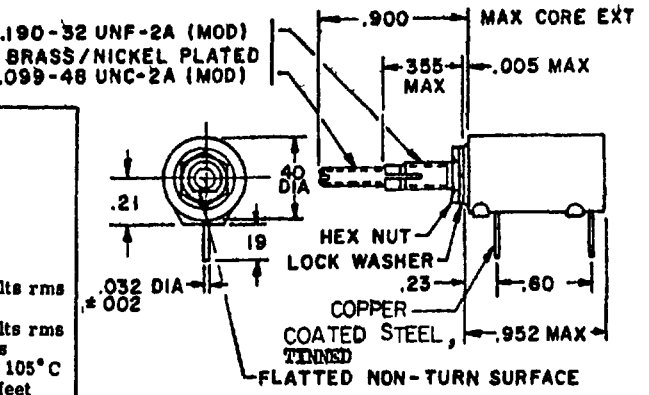
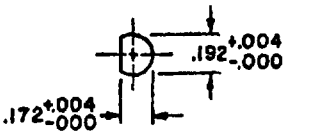
FED. SUP CLASS
5950



RATINGS	
Style	LT12
Grade	3
Class	A
Maximum operating temperature	105°C
Temperature rise	15°C
Ambient temperature	90°C
Dielectric withstanding voltage (sea level)	1000 volts rms
Dielectric withstanding voltage (reduced barometric pressure)	200 volts rms
Terminal pull	5 pounds
Operating temperature	-55° to 105°C
Altitude	70,000 feet

INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.172	4.37	.355	9.02
.004	.10	.19	4.83	.40	10.16
.005	.13	.192	4.88	.60	15.24
.032	.18	.21	5.33	.900	22.86
.099	2.51	.23	5.84	.952	24.18



CHASSIS CUTOUT

ELECTRICAL CHARACTERISTICS (INITIAL)

Dash No 1/	Type designation	Former MS part number	Former type designation	Tuning range						Min self-resonant frequency at max specified inductance	DC resistance max Ohms	Rated DC current ma
				Inductance min	Q min	Test frequency	Inductance max	Q min	Test frequency			
-32	LT12V001	90543-01	LT10V001	.18 µH	85	25 MHz	28 µH	100	25 MHz	225 MHz	.05	1800
-33	LT12V002	90543-02	LT10V002	.25 µH	95	25 MHz	.38 µH	110	25 MHz	200 MHz	055	1720
-34	LT12V003	90543-03	LT10V003	.35 µH	95	25 MHz	.52 µH	110	25 MHz	185 MHz	.06	1650
-35	LT12V004	90543-04	LT10V004	.45 µH	105	25 MHz	.70 µH	110	25 MHz	165 MHz	.08	1430
-36	LT12V005	90543-05	LT10V005	.62 µH	110	25 MHz	1.00 µH	110	25 MHz	125 MHz	.09	1340
-37	LT12V006	90543-06	LT10V006	83 µH	100	25 MHz	1.60 µH	65	7 9 MHz	100 MHz	.10	1280
-38	LT12V007	90543-07	LT10V007	1 30 µH	65	7.9 MHz	2 50 µH	80	7 9 MHz	78 MHz	.23	840
-39	LT12V008	90543-08	LT10V008	1.70 µH	70	7 9 MHz	3 40 µH	85	7.9 MHz	62 MHz	.35	680
-40	LT12V009	90543-09	LT10V009	2 60 µH	70	7 9 MHz	5.50 µH	95	7 9 MHz	52 MHz	.75	465
-41	LT12V010	90543-10	LT10V010	3 80 µH	80	7 9 MHz	7 30 µH	95	7 9 MHz	42 MHz	1 00	405
-42	LT12V011	90543-11	LT10V011	5 80 µH	85	7.9 MHz	11 00 µH	90	2.5 MHz	34 MHz	1 40	340
-43	LT12V012	90543-12	LT10V012	9 30 µH	65	7 9 MHz	16 00 µH	90	2.5 MHz	24 MHz	1.50	315
-44	LT12V013	90543-13	LT10V013	13 50 µH	70	2 5 MHz	27 00 µH	90	2.5 MHz	16 MHz	2 00	274
-45	LT12V014	90543-14	LT10V014	20 00 µH	75	2 5 MHz	40 00 µH	90	2 5 MHz	15 6 MHz	2 20	261
-46	LT12V015	90543-15	LT10V015	30 00 µH	80	2.5 MHz	58.00 µH	90	2.5 MHz	12 5 MHz	3 00	223
-47	LT12V016	90543-16	LT10V016	46 00 µH	80	2 5 MHz	87.00 µH	80	2 5 MHz	10 6 MHz	3 80	198
-48	LT12V017	90543-17	LT10V017	70 00 µH	75	2 5 MHz	130 00 µH	85	790 kHz	8 0 MHz	4 20	189
-49	LT12V018	90543-18	LT10V018	105 00 µH	75	790 kHz	170.00 µH	85	790 kHz	7 0 MHz	4 50	182
-50	LT12V019	90543-19	LT10V019	150 00 µH	65	790 kHz	250.00 µH	80	790 kHz	5 5 MHz	6 00	156
-51	LT12V020	90543-20	LT10V020	225 00 µH	55	790 kHz	390.00 µH	70	790 kHz	3 6 MHz	13 00	107
-52	LT12V021	90543-21	LT10V021	360 00 µH	55	790 kHz	600.00 µH	60	790 kHz	2.8 MHz	17 00	84
-53	LT12V022	90543-22	LT10V022	570.00 µH	35	790 kHz	1 25 mH	45	250 kHz	1 95 MHz	19 00	82
-54	LT12V023	90543-23	LT10V023	1 05 mH	40	250 kHz	2 00 mH	45	250 kHz	1 45 MHz	23 00	74
-55	LT12V024	90543-24	LT10V024	1 60 mH	35	250 kHz	3.20 mH	45	250 kHz	1 3 MHz	33 00	62
-56	LT12V025	90543-25	LT10V025	2 50 mH	35	250 kHz	5 00 mH	40	250 kHz	1 10 MHz	41.00	56
-57	LT12V026	90543-26	LT10V026	3 70 mH	35	250 kHz	7 50 mH	40	250 kHz	.80 MHz	61.00	46
-58	LT12V027	90543-27	LT10V027	5 00 mH	35	250 kHz	10 00 mH	35	250 kHz	65 MHz	81 00	40
-59	LT12V028	90543-28	LT10V028	8 50 mH	35	250 kHz	16 00 mH	35	79 kHz	46 MHz	120 00	32
-60	LT12V029	90543-29	LT10V029	12 50 mH	25	79 kHz	25 00 mH	35	79 kHz	34 MHz	170 00	27
-61	LT12V030	90543-30	LT10V030	20.00 mH	20	79 kHz	38.00 mH	30	79 kHz	28 MHz	230.00	23
-62	LT12V031	90543-31	LT10V031	30 00 mH	20	79 kHz	56 00 mH	30	79 kHz	22 MHz	345 00	19

1/ The dash number added to the MS military standard number constitutes the MS part number. For example, MS 90543-32.

Ⓐ Entire standard revised

APPROVED 24 March 1967 REVISED Ⓐ 27 Nov 1973

P.A EL	International interest	TITLE	MILITARY STANDARD
Other Cust EC 80		COILS, RADIO FREQUENCY, MOLDED, VARIABLE, POWDERED IRON CORE (PRINTED CIRCUIT MT'G) TYPES LT12V001 TO LT12V031 INCLUSIVE	MS 90543
Procurement Specification MIL-C-15305		SUPERSEDES:	PAGE 1 OF 3

DD FORM 1 SEP 63 672 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

5950-0449

User activities: Army MI, ME, EL; Navy AS, CG, MC; Air Force 19

Reviewer activities: Army MI, MU; Navy EC, OS, SH; Air Force 11, 17, 80; DSA-ES

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Electrical characteristics (Final)				
Inspection Group	Allowable variation from initial measurement		Allowable % from specified minimum value in Electrical Characteristics (initial) table	
	Inductance	DC resistance	Self-resonant frequency	Q
	Percent			
Qualification Inspection				
Group II	±2	---	--	-10
Group III	±5	±(2% + .001 ohm)	-5	-15
Group IV	±3	±(2% + .001 ohm)	-5	-5
Quality Conformance Inspection				
Group C Subgroup I	±2	---	--	-10
Subgroup II	±3	±(2% + .001 ohm)	-5	-5
Subgroup III	±5	±(2% + .001 ohm)	-5	-10

NOTES:
1. Core shall be set at the maximum specified inductance value indicated in the electrical characteristics (initial) table prior to all inspection tests. This core setting shall not be changed until electrical characteristics (Final) measurements are performed.
2. Temperature rise and terminal pull tests are applicable.
3. Barometric pressure (test condition C) is applicable.
4. Operating torque shall be a maximum of 6 ounce-inches and a minimum of 0.5 ounce-inch for tuning coil.
5. The test fixture in the diagram below shall be used for electrical measurements. The effective inductance shall be determined for all coils whenever this measurement is specified. Allowance of approximately .03 μH shall be determined for intenal inductance of the Q-meter and test fixture when measuring inductance values of 10 μH or less. The shorting bar shall be made of solid copper wire (AWG #20) bent to conform to the body configuration of the coil under test (see diagram below).
6. Approximate weight:
   Dash number 32-42 -----5.8 grams
   43-52 -----6.6 grams
   53-62 -----7.0 grams
7. Dimensions are in inches.
8. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
9. Unless otherwise specified, tolerances are .XX±.02 and .XXX±.005.
10. For design feature purposes, this standard takes precedence over procurement document referenced herein.
11. Referenced document shall be of the issue in effect on date of invitations for bid.
12. Shock specified pulse method 213, test condition I is applicable. Coil shall be mounted by its mounting stud and securely fastened with mounting hardware for vibration and shock testing.
13. Terminal material shall be in accordance with QQ-W-345, Type I.

APPROVED 24 March 1967 REVISED (A) 27 Nov 1973

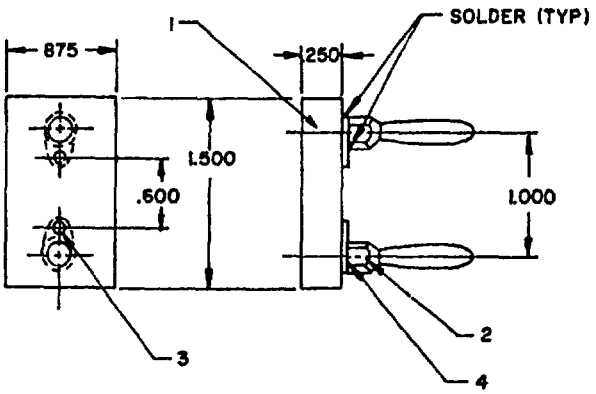
User activities Army MI, ME, EL Navy AS, CG, MC Air Force 19
Reviewer activities Army MI, MU Navy EC, OS, SH Air Force 11, 17, 80 DSA-ES

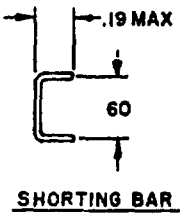
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P.A EL	International Interest	TITLE	MILITARY STANDARD
Other Cust EC 80		COILS, RADIO FREQUENCY, MOLDED, VARIABLE, POWDERED IRON CORE (PRINTED CIRCUIT MT'G) TYPES LT12V001 TO LT12V031 INCLUSIVE	MS 90543
Procurement Specification MIL-C-15305		SUPERSEDES:	PAGE 2 OF 3

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TEST FIXTURE FOR ELECTICAL MEASUREMENTS





Item	Description	Qty
1	Lucite block	1
2	#103 Banana plug or equivalent	2
3	Nugent PS-402-60 or equal	2
4	Copper lug	2

INCHES	MM
.19	4.83
.250	6.35
.600	15.24
.875	22.23
1.000	25.40
1.500	38.10

APPROVED 24 March 1967 REVISED (A) 27 Nov 1973

User activities: Army MI, ME, EL; Navy AS, CG, MC; Air Force 19

Reviewer activities: Army MI, MU; Navy EC, OS, SH; Air Force 11, 17, 80; DSA-ES

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P A EL	International interest	TITLE COILS, RADIO FREQUENCY, MOLDED, VARIABLE, POWDERED IRON CORE (PRINTED CIRCUIT MT'G) TYPES LT12V001 TO LT12V031 INCLUSIVE	MILITARY STANDARD
Other Cust EC 80			MS 90543
Procurement Specification MIL-C-15305		SUPERSEDES:	PAGE 3 OF 3

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION MS90543 COILS,RADIO FREQUENCY,MOLDED,VARIABLE,POWDERED IRON CORE(PRINTED CIRCUIT MT'G)TYPES LT12V001 TO LV12V013 INCLUSIVE

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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